

Operating Manual Micolift M3P

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Chapter 1 Introduction

1.1 Foreword

MICOLIFT M3P Control Systems for Lifts is winning more and more market share after its launch. With beyond-measure security, friendly human-computer interface, and competitive price, MICOLIFT M3P becomes the first-choice for lift alternation and lift reconstruction. It is the mainstream of lift technology development, and thus has been employed by a lot of lift factories in China.

Remark: The working voltage of Micolift M3P is 24VDC. It will damage Micolift M3P if connecting with 220VAC.

1.2 Features

MICOLIFT M3P employs parallel communication. Main processor can handle parallel signals within 8 floors directly. Its main board adopts five-layer board-making techniques and surface-paste technology to maximize system's anti-interference ability. The top and bottom board basically are overlay without wiring. Through friendly human-computer interface, the users do not need any programming; only need to input several parameters according to the real instance. It has the following characteristics.

1.2.1 Hoistway parameters self-learning

1.2.2 Adopting high-speed counting technology and nicety orientation technology to maximize leveling precision.

1.2.3 Calling lifts directly from the system. All the hall call signals and car call signals can be operated and displayed on the system.

1.2.4 Timing automatic closing-door, opening-door and closing-door protection.

1.2.5 Choosing not to answer signals from a certain floor.

1.2.6 Choosing single-door, double-door or not opening-door in a certain floor.

1.2.7 Setting floor display for BCD code, seven-segment code or GeLei code.

1.2.8 Setting base station, and fire-control returning station.

1.2.9 Displaying the pulses-number of every floor after system self-learning, and the location of on-line car.

1.2.10 Can run with parallel connection.

1.2.11 Fifty error history records.

1.2.12 Display running curve.

1.2.13 Supporting remote monitoring or resident section monitor.

With MICOLIFT M3P, you can solve lift problems immediately, and even keep your lifts away from errors in advance.

1.3 Lift Control Functions

Item	Function	Remark
1	Inspection	
2	Full collect control	
3	Self-security run with slow speed	
4	Automatic opening-door on arrival	
5	Door security protection	
6	Orderly hall call press-button operation for opening doors at the current floor	
7	Press-button operation for opening and closing doors	
8	Automatic closing-door time-delay	
9	Automatic set and change direction	
10	Opening doors and orienting through hall call signal	
11	Car call signal record wrong and re-press to cancel	Optional function
12	Automatic cancel command while direction reverse	Optional function
13	Automatic divide speed while single / multi-floor run (over 1.5m/s)	
14	Direct run with full load	
15	Arrival gong	
16	Automatic cut off car light and fan while waiting lift	
17	Automatic return to base station	Activated while parallel or group control. Optional function while single lift
18	Humanity LED operative unit	
19	Communicated with upper position computer	
20	Error history	
21	Hoistway floor data self-learning	
22	Set the sever floor	
23	Set display symbol for floor	
24	Driver operate	
25	Flashed lights of car call to indicate corresponding hall call signal when running with a driver	
26	Automatic answering of car call and cancel decelerating signal	
27	Orderly answering of hall call and cancel decelerating signal	

28	Independent running	
29	Floor displayer of dots matrix	
30	Rolling display of run direction	
31	Automatic correct for floor position signal	
32	Lock lift	
33	Emergency return while fire	Optional for not fire lift
34	Fire man operate	Optional for fire lift
35	Voice report station	Optional function
36	Protection of door safe touch board	
37	Over-loaded alarm and protection	
38	Prevent from disturb of light load	Optional function
39	Protection for run with reverse direction	
40	Protection of proof slip	
41	Stop car by hall call answer of the farthest reverse direction	
42	Constrained speed-changing at the terminal floor	
43	Automatic re-opening-door due to closing-door error	
44	Error protection of inverter	
45	Main control CPU WDT protection	
46	Monitor for village (or mansion)	Optional function
47	Remote monitor	Optional function by modem and telephone line
48	Parallel control run	Optional function
49	Group control run	Optional function
50	Service for rush time while on duty	Optional while group control
51	Waiting machine dispersedly	Activated while group control
52	Display running curve	Must be equipped with tacho encoder (optional function)

Chapter 2 Interface

2.1 Terminal connection with dual speed

Communication I/O functions please find in followed diagram 1.

Table 1 and Table 2 describe every unit's function of Micolift M3P. The working voltage is 24VDC.

Table 1 Input Unit Description

INPUT UNIT		
Terminal	Description	Remark
X00	Third floor call down	NO Contactor
X01	Forth floor call down	NO Contactor
X02	Fifth floor call down	NO Contactor
X03	Sixth floor call down	NO Contactor
X04	Seventh floor call down	NO Contactor
X05	First floor car call	NO Contactor
X06	Second floor car call	NO Contactor
X07	Third floor car call	NO Contactor
X10	Forth floor car call	NO Contactor
X11	Fifth floor car call	NO Contactor
X12	Sixth floor car call	NO Contactor
X13	Seventh floor car call	NO Contactor
X14	Input signal for close door limit position 1	NC Contactor
X15	Fast/Slow run, Monitor input signal for Up/Down direction	NO or NC Contactor (adjustable)
X16	Monitor signal for car door lock	NO Contactor
X17	Monitor signal for hall door lock	NO Contactor

X20	Input signal for fast run inspection	NO Contactor
X21	Monitor signal input for safe circuit	NO Contactor input
X22	Input signal for inspection slow down	NO Contactor
X23	Input signal for inspection slow up	NO Contactor
X24	Input signal for over load	NO or NC Contactor (adjustable)
X25	Input signal for open door limit position 1	NC Contactor
X26	Brake feed back signal	NC Contactor
X27	Input signal for close door pushbutton	NO Contactor
X30	Input signal for safe touch board	NO or NC Contactor (adjustable)
X31	Input signal for open door pushbutton	NO Contactor
X32	Upper leveling	NO Contactor
X33	Inspection / Normal input signal	Inspection status while power off
X34	Input signal for fire protection	NO or NC Contactor (adjustable)
X35	Input signal for lock lift	NO Contactor
X36	Input signal for lower limit position	NC Contactor
X37	Input signal for upper limit position	NC Contactor
X40	Down deceleration	NO Contactor
X41	Up deceleration	NO Contactor
X42	Input signal for second forced change	NC Contactor
X43	Input signal for second upper forced change	NC Contactor
X44	Input signal for open door limit position 2	NC Contactor
X45	Input signal for close door limit position 2	NC Contactor

X46	Input signal for driver	NO Contactor
X47	Input signal for full load	NO Contactor
X50	Down leveling	NO Contactor
X51	Seventh floor call up	NO Contactor
X52	Eighth floor call down	NO Contactor
X53	Eighth floor car call	NO Contactor
X54	First floor call up	NO Contactor
X55	Second floor call up	NO Contactor
X56	Third floor call up	NO Contactor
X57	Forth floor call up	NO Contactor
X60	Fifth floor call up	NO Contactor
X61	Sixth floor call up	NO Contactor
X62	Second floor call down	
VP+	Power for input signal	
A	Close door delay	Close door delay signal while used with dual speed lift
B		
GND	(0V) Power for Input signal	
24V (Left)	(24V) Power for Input signal	For external power
GND	(0V) Power for Input signal	
24V (Right)	(24+) Power for Input signal	For work power

Table 2 Output Unit Description

OUTPUT UNIT		
Terminal	Description	Remark
Y00	Output for vice brake contactor control	Used for open brake while high voltage through break off this signal after 3 seconds brake signal output, and then remain brake open under low voltage.
Y01	Brake output	Drive with main brake contactor
Y02	Output for run contactor control	This point will be empty while used with dual speed lift
Y03	Output for lock lift contactor control	Drived with lock lift contactor loop
Y04	Output for light relay control	Light and fan in the car cabinet
Y05	2# output for close door control	Door operator 2
Y06	2# output for open door control	Door operator 2
Y07	1# output for close door control	Door operator 1
Y10	1# output for open door control	Door operator 1
Y11	Output for run up contactor of lift	
Y12	Output for run down contactor of lift	
Y13	Spare use	
Y14	Output for fast run contactor	
Y15	Output for slow run contactor	
Y16	Output for full reactance contactor	This contactor is short connected reactance while running slow.
Y17	Output for half reactance contactor	This contactor is short connected reactance while running fast.
Y20	Light for first floor call up	
Y21	Light for second floor call up	

Y22	Light for third floor call up	
Y23	Light for forth floor call up	
Y24	Light for second floor call down	
Y25	Light for third floor call down	
Y26	Light for forth floor call down	
Y27	Light for fifth floor call down	
Y30	Light for first floor car call	
Y31	Light for second floor car call	
Y32	Light for third floor car call	
Y33	Light for forth floor car call	
Y34	Light for fifth floor car call	
Y35	Output for buzzer	
Y36	Light for inspection	
Y37	Output for light of run up direction	
Y40	Output for light of run down	
Y41	A	
Y42	B	
Y43	C	
Y44	D	
Y45	E	
Y46	F	
Y47	G	
Y50	-----	Negative output

Y51	Output for fire protection feed back	
Y52	Output for light of over load	
Y53	Output for light of close door delay	
Y54	Light for fifth floor call up	
Y55	Light for sixth floor call up	
Y56	Light for seventh floor call up	
Y57	Light for sixth floor call down	
Y60	Light for seventh floor call down	
Y61	Light for eighth floor call down	
Y62	Light for sixth floor car call	
Y63	Light for seventh floor car call	
Y64	Light for eighth floor car call	

2.2 Terminal connection with VVVF

Communication I/O functions please find in followed diagram 2.

Table 3 and Table 4 describe every unit's function of Microlift M3P. The working voltage is 24VDC.

Table 3 Input Unit Description

INPUT UNIT		
Terminal	Description	Remark
X00	Third floor call down	NO contactor
X01	Forth floor call down	NO contactor
X02	Fifth floor call down	NO contactor
X03	Sixth floor call down	NO contactor

X04	Seventh floor call down	NO contactor
X05	First floor car call	NO contactor
X06	Second floor car call	Inspection down
X07	Third floor car call	Inspection up
X10	Forth floor car call	NO contactor
X11	Fifth floor car call	NO contactor
X12	Sixth floor car call	NO contactor
X13	Seventh floor car call	NO contactor
X14	Input signal for close door limit position 1	NC contactor
X15	Input signal for running feed back	NO or NC Contactor (adjustable)
X16	Monitor signal for car door lock	NO contactor
X17	Monitor signal for hall door lock	NO contactor
X20	Input signal for inverter monitor	NO contactor
X21	Monitor signal input for safe circuit	Input for NO point
X22	Input signal for inspection slow down	NO contactor
X23	Input signal for inspection slow up	NO contactor
X24	Input signal for over load	NO or NC Contactor (adjustable)
X25	Input signal for open door limit position 1	NC Contactor
X26	Brake feed back signal	NC Contactor
X27	Input signal for close door pushbutton	NO contactor
X30	Input signal for safe touch board	NO or NC Contactor (adjustable)
X31	Input signal for open door pushbutton	NO contactor
X32	Input signal for leveling inductor	NO contactor

X33	Inspection / Normal input signal	Inspection status while power off
X34	Input signal for fire protection	NO or NC Contactor (adjustable)
X35	Input signal for lock lift	NO contactor
X36	Input signal for lower limit position	NC Contactor
X37	Input signal for upper limit position	NC Contactor
X40	Input signal for the first down forced change	NC Contactor
X41	Input signal for the first up forced change	NC Contactor
X42	Input signal for the second down forced change	NC Contactor
X43	Input signal for the second up forced change	NC Contactor
X44	Input signal for open door limit position 2	NC Contactor
X45	Input signal for close door limit position 2	NC Contactor
X46	Input signal for driver	NO contactor
X47	Input signal for full load	NO contactor
X50	Input signal for run ready	Used for YASKAWA or FUJI inverter, it is empty while used with Mico inverter. This point will be connected when there is current on inverter.
X51	Seventh floor call up	NO contactor
X52	Eighth floor call down	NO contactor
X53	Eighth floor car call	NO contactor
X54	First floor call up	NO contactor
X55	Second floor call up	NO contactor
X56	Third floor call up	NO contactor
X57	Forth floor call up	NO contactor

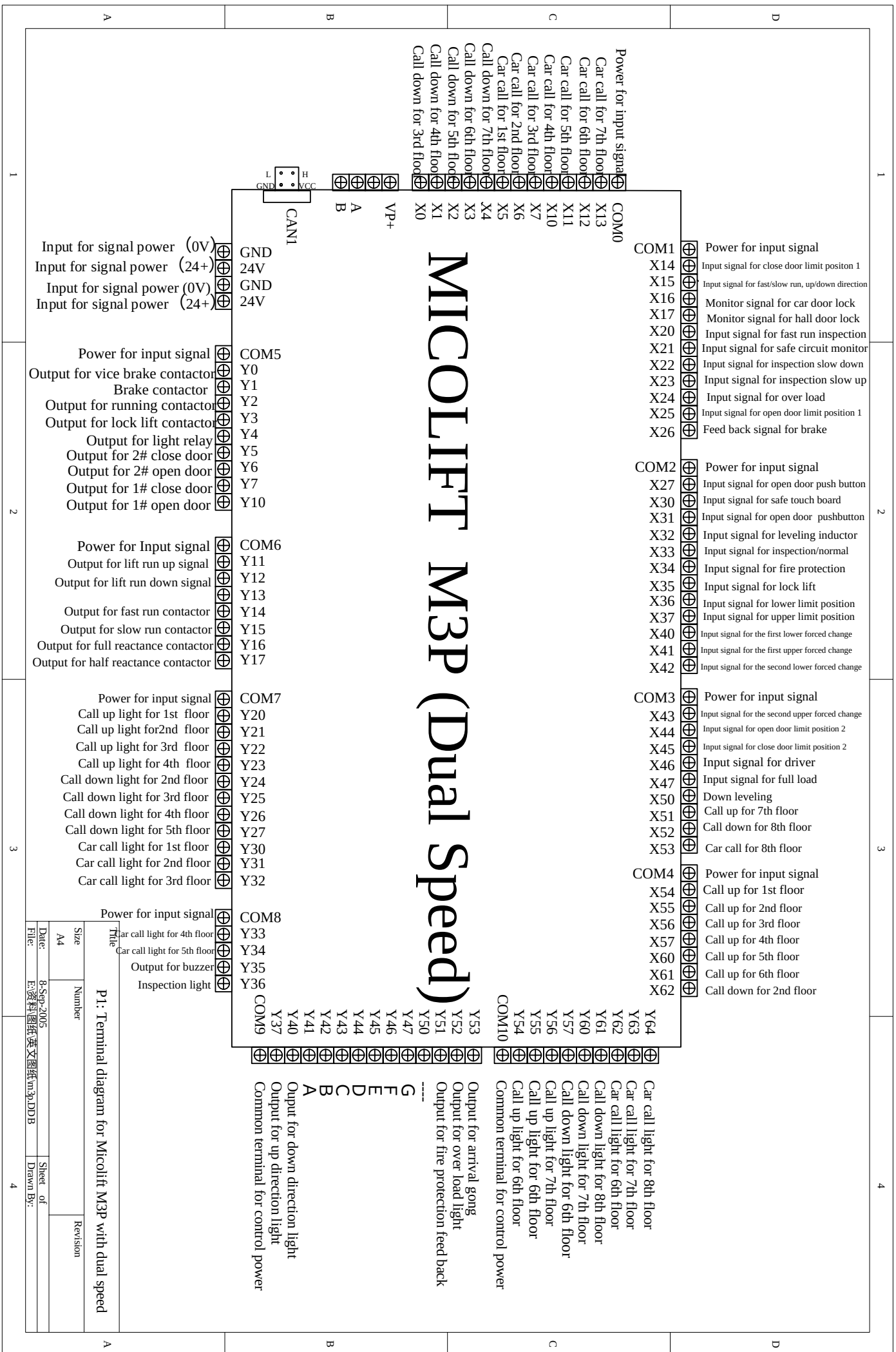
X60	Fifth floor call up	NO contactor
X61	Sixth floor call up	NO contactor
X62	Second floor call down	NO contactor
VP+	Power for input signal	
A	Input for A phase	
B	Input for B phase	
GND	(0V) Power for Input signal	
24V (left)	(24V) Power for Input signal	For external power
GND	(0V) Power for Input signal	
24V (right)	(24V) Power for Input signal	For work power

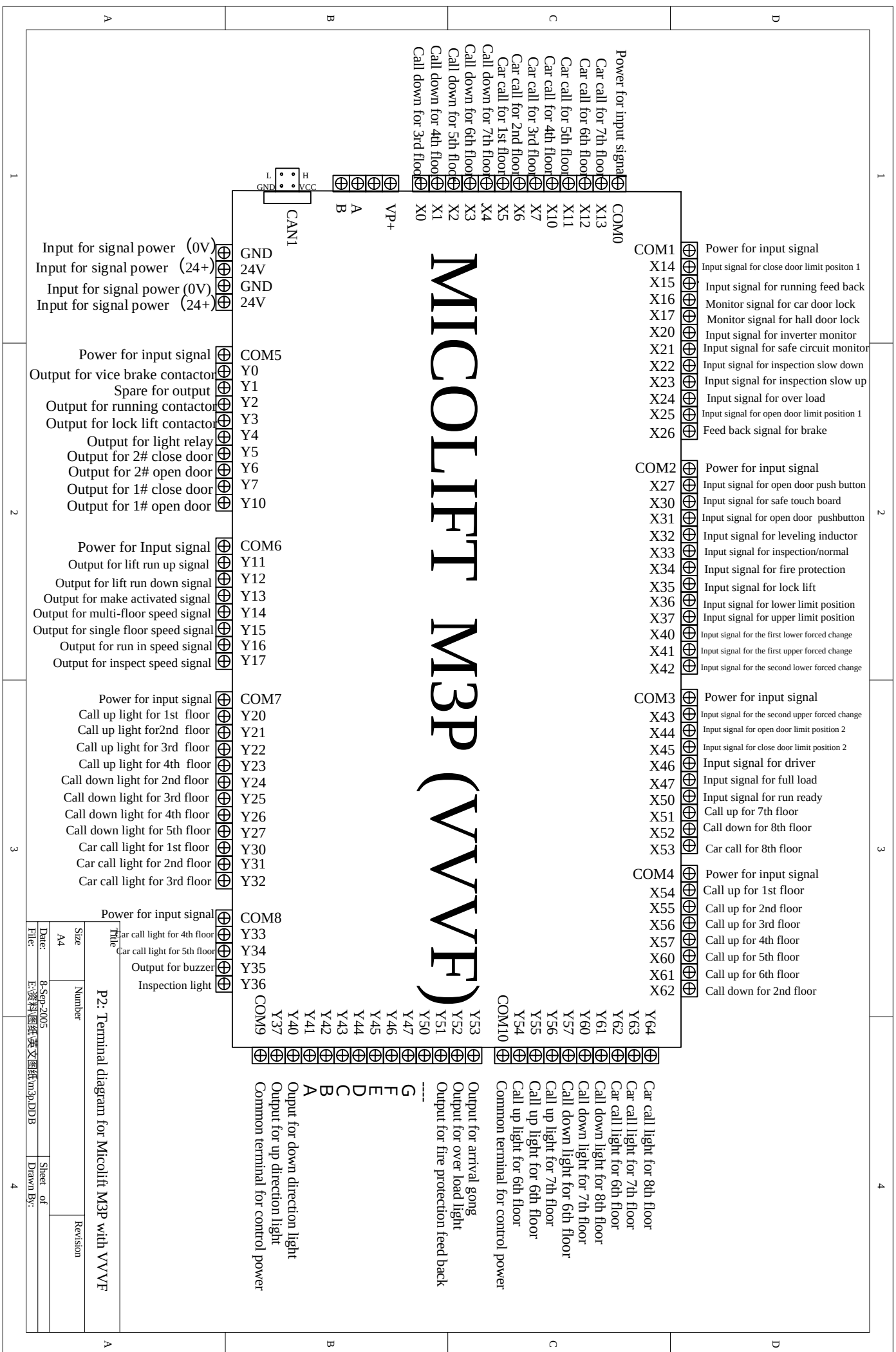
Table 4 Output Unit Description

OUTPUT UNIT		
Terminal	Description	Remark
Y00	Output for vice brake contactor control	Used for open brake while high voltage through break off this signal after 3 seconds brake signal output, and then remain brake open under low voltage.
Y01	Spare use for output	For brake output signal while used with YASKAWA inverter.
Y02	Output for run contactor control	Output running signal while monitor lift signal work normally
Y03	Output for lock lift contactor control	Drived with lock lift contactor loop
Y04	Output for light relay control	Light and fan in the car cabinet
Y05	2# output for close door control	Door operator 2
Y06	2# output for open door control	Door operator 2

Y07	1# output for close door control	Door operator 1
Y10	1# output for open door control	Door operator 1
Y11	Output for run up contactor of lift	
Y12	Output for run down contactor of lift	
Y13	Output for activated signal	Used for KEB inverter
Y14	Output for multi-floor speed signal	Speed command
Y15	Output for single floor speed signal	Speed command
Y16	Output for run in speed signal	Speed command
Y17	Output for inspection speed signal	Speed command
Y20	Light for first floor call up	
Y21	Light for second floor call up	
Y22	Light for third floor call up	
Y23	Light for forth floor call up	
Y24	Light for second floor call down	
Y25	Light for third floor call down	
Y26	Light for forth floor call down	
Y27	Light for fifth floor call down	
Y30	Light for first floor car call	
Y31	Light for second floor car call	
Y32	Light for third floor car call	
Y33	Light for forth floor car call	
Y34	Light for fifth floor car call	
Y35	Output for buzzer	

Y36	Light for inspection	
Y37	Output for light of run up direction	
Y40	Output for light of run down	
Y41	A	
Y42	B	
Y43	C	
Y44	D	
Y45	E	
Y46	F	
Y47	G	
Y50	-----	Negative output
Y51	Output for fire protection feed back	
Y52	Output for light of over load	
Y53	Output for arrival gong	
Y54	Light for fifth floor call up	
Y55	Light for sixth floor call up	
Y56	Light for seventh floor call up	
Y57	Light for sixth floor call down	
Y60	Light for seventh floor call down	
Y61	Light for eighth floor call down	
Y62	Light for sixth floor car call	
Y63	Light for seventh floor car call	
Y64	Light for eighth floor car call	





Chapter 3 Parameter menu for M3P debugger

M3P debugger is used for monitoring and adjusting the parameters. Before adjust the parameters, you should enter password. If it is correct, you can enter adjust interface and the debugger will close the adjust interface if no any key pressed over 15 minutes. You must enter password again if exit the adjust interface. It will turn back to the main interface over 20 minutes and to the homepage over 30 minutes and screen or light behind will be closed over 40 minutes. It has Chinese/English display function and it can be selected through wire.

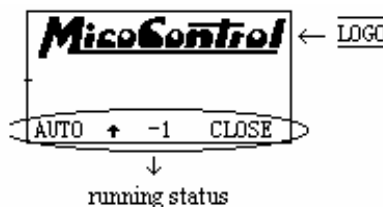
3.1 Screen and the menu

M3P parameter menu debugger is used with LCD of 128X64dot and you can select language Chinese or English through the wire of CE_SET. S1 is used to choose whether the light behind is lighted all long. Selected “Long”, the screen and light behind will be lighted long while selecting “Auto”, it will turn off if no any key pressed. If any key pressed after the screen and light behind closed, it will display again within 1.5 seconds.

First, connect the wires according to the circuit diagram and power on. You can see followed homepage:



Press enter key, enter followed main interface:



The following are the examples of the lift conditions.

AUTO, TEST, FIRE, MANUAL, FULLY, OVER, OPEN, CLOSE, LEARN, LEVELL, LOCK.

In the main interface, press enter key to enter select mode.

Press ↑ and ↓ to select main menu and press Esc key to exit and back to main interface. Press ENTER to enter sub menu.

3.2 Menu Description

MICOLIFT M3P has ten main menu options altogether. Press ↑ and ↓ key to select among them, and press ENTER to enter submenu. The operation is same as the main menu. Under edit mode, press ↑ and ↓ to change parameters and press enter key to save; press esc key to quit. All parameters are set according to default value of factory

but some parameters must be reset according to real conditions.

3.2.1 Main Menu:

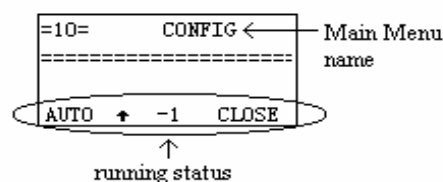
【10】 CONFIG, 【20】 TIME, 【30】 STATION, 【40】 CALL, 【50】 IO MENU, 【60】 DOOR, 【70】 ERROR HISTORY, 【80】 PASSWORD, 【90】 PULSE MONITOR, 【A0】 INPUT SELECT, 【B0】 TIME MENU 2.

Description of the submenus and adjust procedures as followed:

Remark: 【 】 The number in this type symbol stands for the number of main menu.
 《 》 The number in this type symbol stands for the number of sub-menu.

3.2.2 【10】 CONFIG Menu

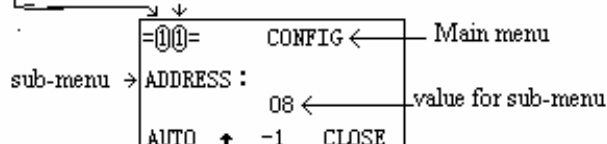
Under first main menu—CONFIG:



Press enter key to enter first sub-menu (Address) of CONFIG menu.

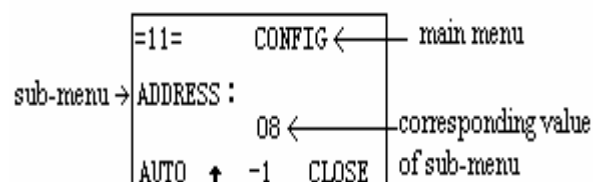
This digital stands for which main menu and "1" stands for first main menu

This digital stands for which sub-menu and "1" stands for first sub-menu.

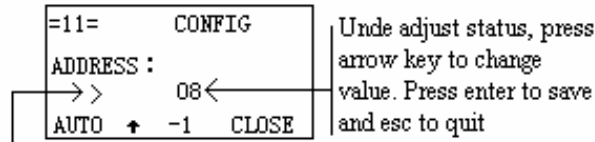


Description for sub menu:

- ✧ 《11》 ADDRESS: range from 0 to 15, setting address of the system in the case of parallel-control, group-controlling or remote monitoring. While two lifts parallel connecting, set one to 1 and the other to 2. It'd better to shut off the power again after setting address.

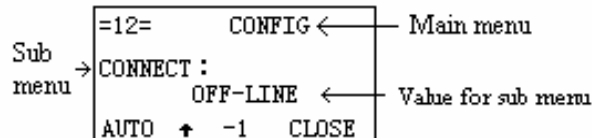


Press enter key to adjust the value for address and it will display ">" on the left of the parameter.

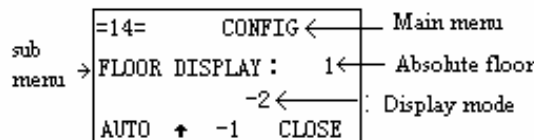


This signal shows it is under adjust status.

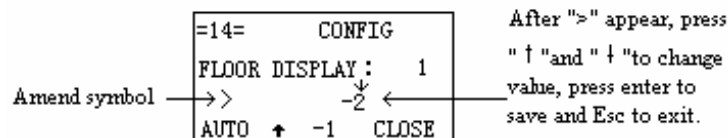
- ✧ 《12》 CONNECT: Choosing the OFF-LINE mode, for spare use. (adjust the parameters same as above)



- ✧ 《13》 SPEED: Input the lift's rating speed. If $V < 1.5\text{m/s}$, the system outputs a high-speed signal; if $V \geq 1.6\text{m/s}$, the system outputs running curve depending on single-floor or multi-floor and it runs fast by two speeds. If $V > 2.0\text{m/s}$, it runs fast with three speed and it can over-floor through decelerating point.
- ✧ 《14》 FLOOR DISPLAY: set up the floor display manner. Press ENTER to enter the submenu.



The numerical value of the absolute floor is displayed on the top right corner, such as "1, 2, 3.....64". The number in the middle is which needed to show. If the absolute floor is the first floor, and "-2" floor will be displayed. To adjust the display mode, press ENTER, then amendment mouse ">" is highlighted. Press ARROW key to adjust the value and press ENTER to save; ESC to exit.



Letter-display is also provided; if some letters are not with the system, please contact with us.

- ✧ 《17》 LEARNING: Set self-learning function. It will turn to automatic run status after lift return to lower position station and enter into door zone (i.e. position station lower forced switch off and door zone connected). It will begin to self learn

after closing door under ON mode. The self learning will finish and lift stop automatically while lift run to upper position station when reaching door zone.

Note: The self learning is successful while the floor value increased sequential. The floor value can be 3 maximum if connecting direction wrong for phase A-B or without pulse input. For dual speed lift, it needn't self-learning just only to enter the highest floor number.

- ✧ 《18》 MAUNAL DOOR: Under manual status, it need to press pushbutton to close door for long if set to ON. If set to OFF, the lift will stop by hall call signal.
- ✧ 《19》 MANUAL DIRECTION: While landing call under manual status, it cannot stop lift while set to “ON”. If set to “OFF”, the lift can be stopped under landing call conditions by manual status.
- ✧ 《1A》 CONVERTER SELECT: Select frequency inverter.
- ✧ 《1B》 DIFERFLOOR: Difference between the floor absolute value of two lifts while two lifts parallel connection. While the floor of two lifts is same, this value is “0”; and while one lift has base floor and the other doesn't, the value will be “1”. MicoM3 micro controller is defined that address of lift without base floor (《11》 address menu) is “2” and the other is “1”.
- ✧ 《1C》 Software version.
- ✧ 《1D》 Pulse number for encoder. If the pulse divided, it needs to enter the pulse number after frequency divided. If enter wrong, the running curve will be possible display not normally.

3.2.3 TIME Menu (【20】 Time)

- ✧ 《21》 STOP: Set the delay time for main contactor off when all speed signals deleted. If using YASKAWA inverter, it is set for brake off. If using dual speed lift, this menu is spare for use.

=21=	TIME
STOP :	0002.0 /S
AUTO	↑ -1 CLOSE

Press enter key to enter in amend status and press arrow key to change the value. Then press enter key to save and ESC to quit.

- ✧ 《22》 START: set the time to open increasing curve. It is used for YASKAWA / FUJI inverter. It is spare for use of dual speed lift.
- ✧ 《23》 BRAKE: set the time to open the brake. It is used for YASKAWA / FUJI inverter. It is spare for use of dual speed lift.
- ✧ 《24》 DOOR OPEN: set the time to open door in advance. For VVVF lift, the time is calculated after the lift decelerates and all speed signals break. For dual speed lift, it is calculated after the brake opened.
- ✧ 《25》 DOOR CLOSE: set the time to close door, showing in seconds.

- ✧ 《26》 OPEN PROTECT: set the time of door-open protect. When door-open limit switch cannot be shut off, this setting can stop opening to avoid the danger of electrifying the door too long.
- ✧ 《27》 CLOSE PROTECT: set the time of door-close protect. When door-close limit switch or door lock error happens, this setting can stop closing and re-open the door.
- ✧ 《28》 RINGING: set the alarm ringing times when receiving hall call signal. This setting is used in the MANUAL mode.
- ✧ 《29》 GONG: set the lasting time of arrival gong.
- ✧ 《2A》 SPEED STOP: Set the delay time to cut off all speed signals while entering into door zone. It will cut off all speed signals when the pulse value reach to set value after decelerating to door zone. In case of pulse calculate invalidate, this time is set to protect so it will later than the time normally cut off. But make sure this time doesn't too long otherwise it cannot stop to nearest station while self secure.
- ✧ 《2B》 FLOOR PROTECT: Floor protect time. The micro controller must get decelerate point signal of each floor in this time; otherwise the micro controller will display E4 error and the lift will decelerate and stop at the nearest.
- ✧ 《2C》 TIME1: While adjusting the inspection speed, delay the time for brake direction after the brake off. When the value is of "0", it won't delay.
- ✧ 《2D》 TIME2: This is for setting time for protecting steel wire rope slip while the car standstill. If the leveling inductor without change within this limited time, it will display E11 of running over time error. If this error occurs, it must shut off the power or open the inspection switch to reset.
- ✧ 《2E》 TIME3: Spare.
- ✧ 《2F》 RUN TIME: The running times of the lift. It just calculates the running times while the lift run fast.

3.2.4 STATION Menu (【30】 Station)

- ✧ 《31》 BASE: set the base station and locked-floor station of the parallel connected lift. Setting value of the base station accords to the absolute floor value.

=31=	STATION
BASE :	01
AUTO	↑ -1 CLOSE

Press enter key to enter in amend status and press arrow key to change the value. Then press enter key to save and ESC to quit.

- ✧ 《32》 FIRE HOUSE: set the returning floor value when the lift under the condition of fire-protect. This value must be set according to the absolute floor value.
- ✧ 《33》 HIGHEST: set the highest floor of the lift, according to the result of the system automatic test.
- ✧ 《34》 WAIT(1): set the waiting floor value when the lift under the condition of group-control or parallel-connected.
- ✧ 《35》 WAIT(2): set the waiting floor when the lift under the condition of

group-control or parallel-connected.

- ✧ 《36》 REPLY STATION: set the replying floor value.

36 REPLY STATION															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
AUTO		1													

It is displayed with binary digits, “1” representing replying and “0” for no reply. From left to right is the floor value of “1, 2....., 64”.

display the floor which highlight

the location which mouse highlighted →

36 REPLY STATION																(01)
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
AUTO		1														

← "1" stands for the replying floor.
← "0" stands for 48 floor which no reply.

Under select floor mode, press arrow key to select floor.

After pressing ENTER, one of the floor value is highlighted with the mouse in the right top of the screen to amendment. Press arrow key to change the location of mouse and press enter key to amend the reply status of corresponding floor (the amend symbol will show on the left). Press ↑ key to set “0” then press ↓ key to set “1”.

display the floor highlighted

high light ed mou se →

36 REPLY STATION																> (01)
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
AUTO		1														

amend symbol. press arrow key to change the value.

With this function, the reply station can be set without wiring.

3.2.5 CALL Menu (【40】 Call)

This menu can observe or login calling lift signals of every floor and monitor and amend the hall call signals.

- ✧ 《41》 UP CALL: it can observe or login the calling lift signals below 64 floors and show with the binary digits. “1” representing calling-lift signal and “0” for no signal.

menu sub-menu
num. ↘ ↓

=51= INPUT1																
A	b	X00	X01	X02	X03	X04	X05									
X06	X07	X10	X11	X12	X13	X14	X15									
X16	X17	X20	X21	X22	X23	X24	X25									
X26	X27	X30	X31	X32	X33											
TEST											1					

input
terminal

If there is signal input, the name of terminal will be reverse video; if no, there is no signal input.

This terminal is high lighted and signal input.

=51= INPUT1																
A	b	X00	X01	X02	X03	X04	X05									
X06	X07	X10	X11	X12	X13	X14	X15									
X16	X17	X20	X21	X22	X23	X24	X25									
X26	X27	X30	X31	X32	X33											
TEST											1					

input
terminal

These corresponding terminal without signal input.

The code accords to the code of input signal. For example, “A-B” stands pulse input A-B, “X00” for X00 terminal, “X01” representing X01 terminal and so on.

《52》 INPUT 2: Display the status of system’s parallel input units. The code accords to the code of input signal.

= 52 = INPUT2																
X34	X35	X36	X37	X40	X41	X42	X43									
X44	X45	X46	X47	X50	X51	X52	X53									
X54	X55	X56	X57	X60	X61	X62	X63									
AUTO											1					

✧ 《53》 OUTPUT 1: Display the status of system’s parallel output units, used as INPUT.

= 53 = OUTPUT1																
Y00	Y01	Y02	Y03	Y04	Y05	Y06	Y07									
Y10	X11	Y12	Y13	Y14	Y15	Y16	Y17									
Y20	Y21	Y22	Y23	Y24	Y25	Y26	Y27									
AUTO											1					

There is signal output of terminal

The code accords to the code of output signal. For example, “Y00” stands for Y00 terminal, “Y01” representing Y01 terminal, “Y11” for Y11 terminal and so on.

✧ 《54》 OUTPUT 2: Display the status of system’s parallel output units. The code accords to the code of output signal.

```

= 54 = OUTPUT2
Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37
Y40 Y41 Y42 Y43 Y44 Y45 Y46 Y47
Y50 Y51 Y52 Y53 Y54 Y55 Y56 Y57
Y60 Y61 Y62 Y63 Y64
AUTO      1

```

3.2.7 DOOR Menu (【60】 Door)

- ✧ 《61》 DOOR 1: set the floor value of opening the lift door of door machine 1# represented by binary digits, with “1” representing open and “0” for close.

"1" stands for opening door of corresponding floor.(1-32)

```

=61 DOOR ( 1 )
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AUTO      1

```

"0" stands for closing door for corresponding floor.(33-64)

Press ENTER, then the highlighted floor value is displayed in the right top of the screen for amendment.

Press up and down arrow key to change the mouse location.

```

display floor high lighted
=61= DOOR ( 1 ) ( 01 )
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AUTO      1

```

Press enter key again, the amend symbol will appear and press ↑ and ↓ to change the location of high lighter mouse.

display floor high lighted
amend symbol

```

=61= DOOR ( 1 ) > ( 01 )
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AUTO      1

```

Press ↓ to change the value to "1"

Press ↑ change value to "0" while the mouse high lighted.

- ✧ 《62》 DOOR 2: set floor value for opening door of door machine 2#. Used as DOOR 1.

Note: If setting one floor no reply (set in station menu), door 1 and door2 must be set to close door.

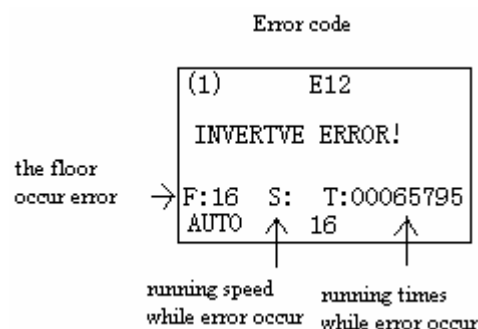
3.2.8 ERROR HISTORY Menu (【70】 Error History)

When error occurs, the code and cause of the error are displayed in the bottom of screen. (Floor display has corresponding code too please see the detail in Chapter 4.)



Error occurred, the error code and cause will rolling display.

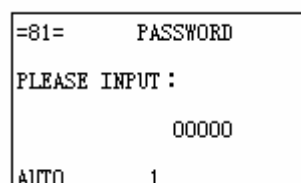
MICOLIFT M3 system records maximum 50 history errors. Through this menu, user can check which floor occurs error, the running speed of that time and the running times of lift.



Checking error history, press enter key to enter dele mode and you can dele all error histories after two times dele confirmation.

3.2.9 PASSWORD Menu (【80】 Password)

The password is represented by five digits.



All the operations are validated by the correct password. The default password is 00000. After inputting the password, if the power is reset or no any key be pressed within 15 minutes, the password must be re-enter to activate the operations. If enter correct password again, it will enter to the status to amend password. The new password will be saved after confirm the amendment.

Note: If the password is lost, all the amendment operations can not be used. The system must be decrypted by us.

3.2.10 PULSE MONITOR Menu (【90】 Pulse monitor, while used with dual

speed lift, this menu is just spare for use.)

- ✧ 《91》 CURRENT (current pulse): display the current position of the lift-car with pulse number. This parameter is just can be checked and not to be amended.

```

=91=    PULSE MONITOR
CURRENT :
        00000000
AUTO    1

```

Remark: Under this menu, you can enter into running curve interface by pressing “enter” key.

- ✧ 《92》 RUN IN (leveling pulse): length of door zone. 1/4 length of magnetism proof board. Amend this value to change the length of door zone corresponding pulse number.
- ✧ 《93》 DEL. 1 DISTANCE: deceleration distance 1, change the value to adjust the length of deceleration distance.
- ✧ 《94》 DEL. 2 DISTANCE: deceleration distance 2, change the value to adjust the length of deceleration distance.
- ✧ 《95》 DEL. 3 DISTANCE: deceleration distance 3, change the value to adjust the length of deceleration distance.
- ✧ 《96》 FLOOR: display the pulse number of each floor. Press enter key to check the pulse number.

```

                floor displayed
                ┌──────────┐
=96=    PULSE MONITOR
FLOOR :    01 ←┘
        00000000 ←┘
AUTO    1
                └──────────┘
                pulse number

```

Press ↑ or ↓ key to check pulse number of 1-64 floors. Press enter key to change the pulse number.

```

=96=    PULSE MONITOR
FLOOR :    01
        > 00000000
AUTO    1

```

Under amend status, press ↑ or ↓ key to change pulse number value. Press enter to save and esc to quit.

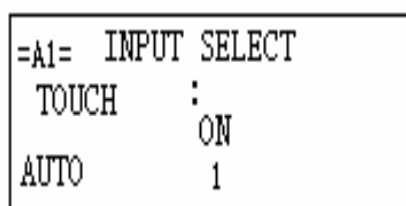
- ✧ 《97》 V2 DEL PERMIT (XDL): Set the distance from start to accelerate to medium speed while selecting the single/multi-floor run curve according to the distance. This value is multiple of length of magnetism proof board. Select the single/multi-floors run curve depends on floor and this menu is spare.
- ✧ 《98》 V3 DEL PERMIT (XDL): Set the distance from start to accelerate to high

speed while selecting the single/multi-floor run curve according to the distance. This value is multiple of length of magnetism proof board. Select the single/multi-floors run curve depends on floor and this menu is spare.

- ✧ 《99》 RUN V2 DEL PERMIT (XDL): Set the distance for running medium speed while selecting the single/multi-floor run curve according to the distance. This value is multiple of length of magnetism proof board. Select the single/multi-floors run curve depends on floor and this menu is spare.
- ✧ 《9A》 RUN V3 DEL PERMIT (XDL): Set the distance for running high speed while selecting the single/multi-floor run curve according to the distance. This value is multiple of length of magnetism proof board. Select the single/multi-floors run curve depends on floor and this menu is spare.
- ✧ 《9B》 DOOR LENGTH: length of door zone.

With this monitor function, the data of lift self-learning can be examined. Also the precision of leveling can be adjusted. For example, when the lift runs up or down to the second floor, the lift-car is lower than the leveling point. This problem can be solved by increasing the pulse number of the second floor until the lift levels precisely.

3.2.11 INPUT SELECT MENU (【A0】 Input select)



- ✧ 《A1》 TOUCH: Select NO or NC for safe touch board signal. (ON for NO, OFF for NC.)
- ✧ 《A2》 BRAKE: Select NO or NC for brake feed back signal. (ON for NO, OFF for NC.)
- ✧ 《A3》 FIRE: Select NO or NC for first signal. (ON for NO, OFF for NC.)
- ✧ 《A4》 CONTACT: Select NO or NC for contactor feed back signal. (ON for NO, OFF for NC.)
- ✧ 《A5》 OVERLOAD: Select NO or NC for over load signal. (ON for NO, OFF for NC.)
- ✧ 《A6》 SWITCH DOOR LIMIT: Select NO or NC contactor for open/close door limit position. Normally, this function is forbidden and the open / close door limit position is used for NC contactor.

3.2.12 TIME MENU 2 (【B0】 TIME 2) (This menu is for dual speed lift and it is spare while used with frequency inverter.)

- ✧ 《B1》 SOON TIME: Fast Accelerate Speed, to set the delay time of 1A switch on.
- ✧ 《B2》 SLOW TIME: Slow accelerate speed: to set delay time of 2A switch on after

1A switch on again.

- ✧ 《B3》 BRAKE: Set delay time for 1A switch on again after 1A break off.
- ✧ 《B4》 CLOSE LAMP: Set the time for close light and calculated by minutes.
- ✧ 《B5》 CLOSE DOOR DELAY: Set the delay time for close door and calculated by minutes.

3.3 Self-learning of the hoistway data (used for frequency inverter. If for dual speed lift, it needn't self-learning just only input the highest floor.)

- ✧ Make sure all the hoistway switches are in order, such as limit switch, forced decelerating switch, and leveling switch.
- ✧ Adjust ascending and descending forced deceleration distance.

While inspection, it runs to the lower station. I.e. the lower forced switch of terminal station is off and the leveling switch connecting.

- ✧ Lift is automatic running after return back to station.
- ✧ Enter into the sub menu learning of config menu to select ON mode after automatically close door. The lift starts running in the inspect speed automatically. It automatically stops when arriving the top floor level. After finishing self-learning of the hoistway data, the lift can run in high speed.

Note: to optimize the use of the lift, self-learn the hoistway data after re-adjusting the forced deceleration distance or re-adjusting the position signal of door zone.

Some notices during self learning:

- 1, To sure self learning successful, the leveling sensor and forced switch must be correct.
- 2, Make sure the input of pulse encoder must be correct and adopting two phase counting. The connection for pulse input and phase must be accurate in order that pulse will be increased while running up and decreased while running down. Otherwise, please change the A-B phase of input. (Note: the A-B phase of encoder input cannot be changed.)
- 3, Self learning is successful if the floor display changes and stops until increasing to the highest floor. Otherwise, you should check whether the leveling inductor, forced switch and input and phase of encoder pulse is correct.

3.4 Adjusting Leveling Precision

(1) Test method for VVVF lift

To improve leveling precision and speed up inspection process, during the course of installing the magnetism-proof board, let the inductor in the middle of the board while the lift-car at the leveling position. The magnetism-proof board is about 200 – 250 mm in length.

MICOLIFT M3 system provides a function of adjusting the floor pulse number to

adjust leveling precision, needn't entering the hoistway to adjust the magnetism-proof board, thus speed up inspection process. The following are adjusting steps.

- ✧ During installing the magnetism-proof board, let the inductor in the middle of the board while the lift-car at the leveling position.
- ✧ Adjust the inverter to move the lift to a certain floor upwards and downwards, stop at the same position.
- ✧ Comparing the real leveling position and the ideal leveling point, if the lift-car doesn't arrive at the ideal leveling point, then increase the pulse number of this floor; otherwise decrease the pulse number of the floor, until the lift levels precisely.
- ✧ To adjust leveling through amend the pulse number. As the leveling will change after self-learning again, it is advised to reach leveling through moving leveling inductor. Method is as followed:

Adjust the inverter to move the lift to a certain floor upwards and downwards, stop at the same position.

Measure the differ value ΔY between real position and leveling point. If the leveling position is lower than leveling point, it should move the leveling inductor upward of the distance for ΔY . Contrarily, it should move the leveling inductor downward of the distance for ΔY .

- ✧ Test method for dual speed lift:
Just move upward or downward properly for the leveling inductor according to the leveling point and used with brake to test when necessary.

3.5 Establishing the deceleration distance

The system establishes the ascending and descending braking points according to the value of the descending forced braking distance. This value is automatically measured during the course of hoistway data measurement; it can also be adjusted by the user. For lifts running more than 1.6m/s, two forced braking switch are installed for ascending use and descending use. The system will measured low-speed forced braking distance (the second lower forced braking distance) as every floor's low-speed braking distance and first lower forced braking distance as high speed braking distance while self-learning hoistway data. For lifts of 2.0m/s, high-speed braking distance is twice of the medium-speed braking distance. According to the above principles, the medium speed forced braking distance must be proper and self learn hoistway data again.

The reference value of the forced deceleration distance

For lifts of 1.0m/s: 1.5m – 1.7m;

For lifts of 1.5m/s: 2.3m – 2.6m;

For lifts of 1.75m/s: second forced deceleration distance: 1.5m – 1.8m;

first forced deceleration distance: 2.5m – 3.0m;

For lifts of 2.0m/s: second forced deceleration distance: 1.5m – 1.7m;

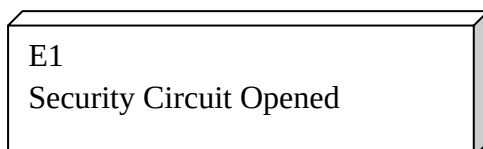
first forced deceleration distance: 3.0m – 3.4m.

Chapter 4 Error Description

MICOLIFT M3P system records 50 errors, with the happening time and the causes. When error happens, the error cause is displayed on the system LCD screen in English and is displayed on the floor display with matching error code.

The followings are the error prompts and the solution.

✧ Error 1



When this error occurs, it will display “E” and “1” alternately for 3 times and display current floor for few seconds and then display “E” and “1” alternately again; such circularly.

Security Circuit Opened. Check the security circuit. The error prompt disappeared after safe circuit work normally.

✧ Error 2



When this error occurs, it will display “E” and “2” alternately for 3 times and display current floor for few seconds and then display “E” and “2” alternately again; such circularly.

Door Lock Opened or Error. Check the door lock and lock circuit. The error prompt disappeared after connecting door lock or inspection.

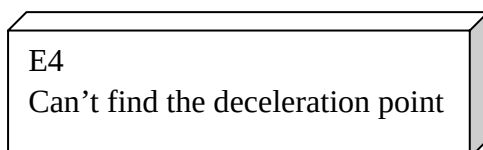
✧ Error 3



When this error occurs, it will display “E” and “3” alternately for 3 times and display current floor for few seconds and then display “E” and “3” alternately again; such circularly.

Can't Find the Leveling Point. Check the leveling inductor and the wires. The lift runs at run in speed until receiving the door zone signal and the lift stops and opens the door. The error prompt disappeared after solving the problem or run inspection.

✧ Error 4



When this error occurs, it will display “E” and “4” alternately for 3 times and display current floor for few seconds and then display “E” and “4” alternately again; such

circularly.

Can't Find the Deceleration Point. Check whether the protection time for floor in 2B menu is right and it should increase when necessary. Check the frequency-separator. The error prompt disappeared when decelerating to door zone. If defeated, please replace parts.

✧ Error 5



When this error occurs, it will display "E" and "5" alternately for 3 times and display current floor for few seconds and then display "E" and "5" alternately again; such circularly.

Up Limited Switch Opened. Check the wires. The error prompt disappeared after connecting up limited switch. If lift reaches up limited position, it can run downward.

✧ Error 6



When this error occurs, it will display "E" and "6" alternately for 3 times and display current floor for few seconds and then display "E" and "6" alternately again; such circularly.

Down Limited Switch Opened. Check the wires. The error prompt disappeared after connecting down limited switch. If lift reaches up limited position, it can run upward.

✧ Error 7



When this error occurs, it will display "E" and "7" alternately for 3 times and display current floor for few seconds and then display "E" and "7" alternately again; such circularly.

Deceleration Switch Error. The lift will stop suddenly if single floor forced change without the deceleration signal from micro processor and leveling with self-survive speed and record E7 error after the lift runs with multi-floor speed.

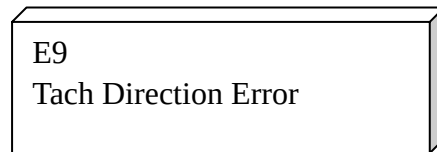
✧ Error 8



When this error occurs, it will display "E" and "8" alternately for 3 times and display current floor for few seconds and then display "E" and "8" alternately again; such circularly.

Contact Not Release. Main contact or braking-contact has not been released when without signal output. Check the wires or replace. The error prompt disappeared after solving the problem.

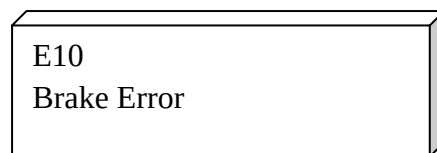
✧ Error 9



When this error occurs, it will display “E” and “9” alternately for 3 times and display current floor for few seconds and then display “E” and “9” alternately again; such circularly.

The input direction of encoder is wrong. If error occurs, please change the direction A-B phase of encoder. Note: It is forbidden to change the input value of A/B phase of frequency inverter.

✧ Error 10



When this error occurs, it will display “1” and “0” alternately for 3 times and display current floor for few seconds and then display “1” and “0” alternately again; such circularly.

Brake error. The brake contactor closed but the mechanical brake of motor is opened or the brake is opened but no signal. Check the brake connection.

✧ Error 11



When this error occurs, it will display “1” and “1” alternately for 3 times and display current floor for few seconds and then display “1” and “1” alternately again; such circularly.

Normally, the lift will stop urgently if run time over the total run time set in micro-processor. The error will be disappeared after inspection. If the signal of leveling inductor doesn't change, Error 11 will display.

✧ Error 12

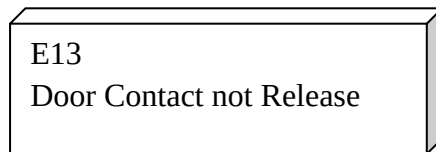


When this error occurs, it will display “1” and “2” alternately for 3 times and display current floor for few seconds and then display “1” and “2” alternately again; such

circularly.

Inverter Error. Check the inverter. This error will be displayed when the inspection signal of inverter off.

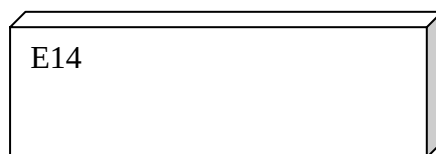
✧ Error 13



When this error occurs, it will display “1” and “3” alternately for 3 times and display current floor for few seconds and then display “1” and “3” alternately again; such circularly.

Door contactor is not released.

✧ Error 14

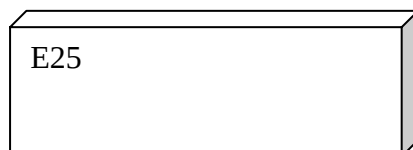


When this error occurs, it will display “1” and “4” alternately for 3 times and display current floor for few seconds and then display “1” and “4” alternately again; such circularly.

✧ Error 15



✧ Error 25



When this error occurs, it will display “2” and “5” alternately for 3 times and display current floor for few seconds and then display “2” and “5” alternately again; such circularly.

REMARK:

While using with BCD code output for micro processor, the error code will display “0” and “X” instead of “E” and “X”. (“X” stands for error code.)

All the types and performances may be changed and won't notify specially.